Machine Learning

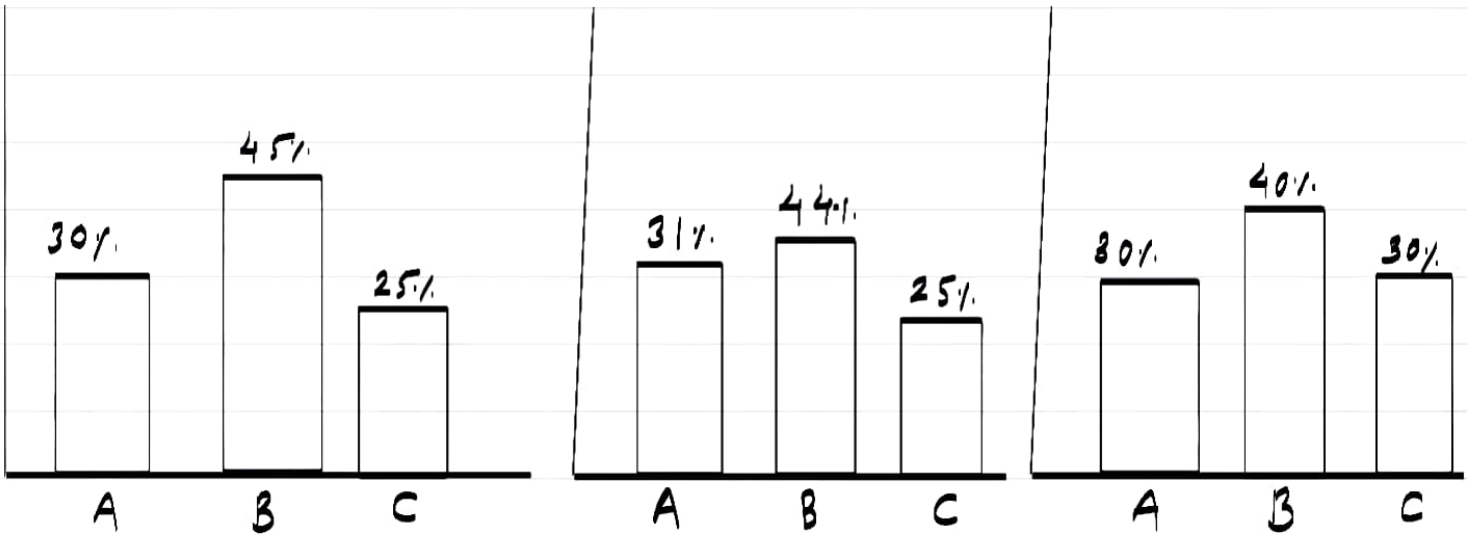
24-03-26

Train test split

categorical data

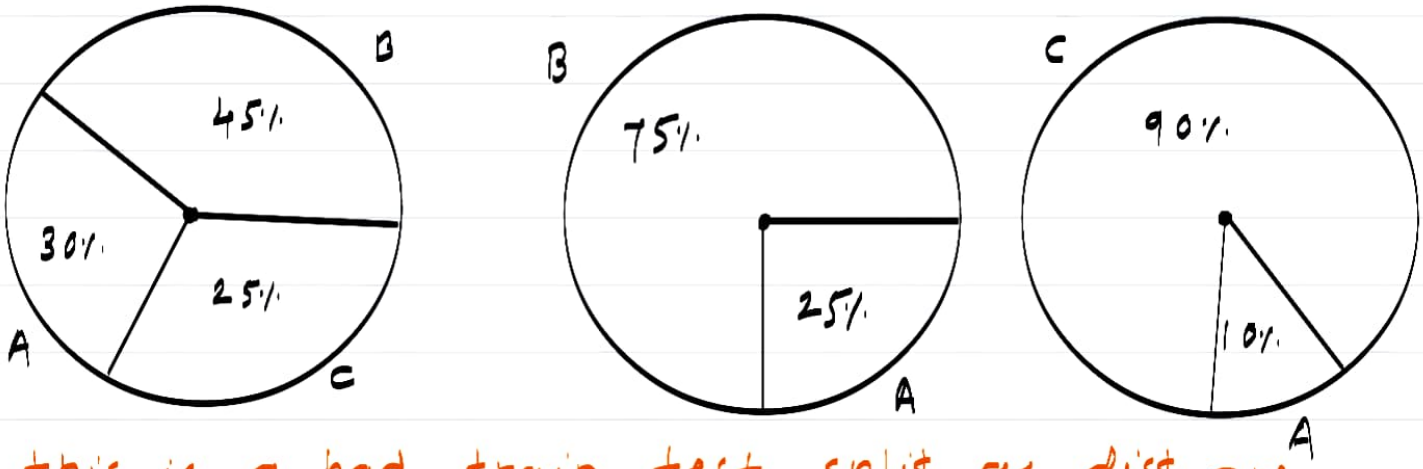
eg. feature - city

available data | train | test



- this is good split as dist. of city categorical data is nearly same in available - train - test datasets

available | train | test



- this is a bad train test split as dist. are very different among subsets

- We use simple random sampling in most of the cases.
  Simple random sampling keeps the distributions similar

- In some cases, when target is categorical data, stratified sampling can be also used.

eg. 1000 rows for fruit classification

target → type of fruit
A B C

available data		train	test
A - 50%	simple random sampling →	A - 48%	A - 55%
B - 25%		B - 22%	B - 25%
C - 25%		C - 30%	C - 20%
		approx proportion match	
	stratified sampling →	A - 50%	A - 50%
		B - 25%	B - 25%
		C - 25%	C - 25%
		exact proportion match	

KNN Regression

steps involved in training

1) Train test split
2) store train data into model
3) set a value of K

steps involved in Evaluation

A) compute performance on train data for every datapoint

1) get its vector representation from data

$$V_1 = [\ 1 \quad 0.5 \quad 0.2 \quad 1.5\ ]$$

$$\begin{array}{cccc} \downarrow & \downarrow & \downarrow & \downarrow \\ \text{area} & \text{rooms} & \text{floor} & \text{parking area} \end{array} \rightarrow \text{features}$$

2) compute its euclidean distance from rest of train datapoints

$$V_i = [\ a_1 \quad a_2 \quad a_3 \quad a_4\ ]$$

$$V_{100} = [\ 0.5 \quad 0.5 \quad 0.3 \quad 1\ ]$$

$$\text{distance}(V_1, V_{100}) = \sqrt{(1-0.5)^2 + (0.5-0.5)^2 + (0.2-0.3)^2 + (1.5-1)^2}$$

$$= (0.25 + 0 + 0.01 + 0.25)^{1/2}$$

$$= (0.51)^{1/2}$$

$$\begin{array}{ccccccc} V_1 \rightarrow & V_2 & V_3 & V_4 & V_5 \cdots\cdots & V_n & \text{vectors} \\ & \downarrow & \downarrow & \downarrow & \downarrow & \downarrow & \\ & d_2 & d_3 & d_4 & d_5 & d_n & \text{distances} \end{array}$$

3) order distances in ascending order and pick 'K' nearest vectors with least distance

lets say $K = 3$ we found nearest vectors as V_{100} V_{20} V_{17}

4) find the average of target values of nearest neighbors

	v_{100}	v_{20}	v_{17}
price	1.5	1.2	0.9

for v_1, model prediction is

$$\frac{1.5 + 1.2 + 0.9}{3} = \frac{3.6}{3} = 1.2$$

v_1 predicted price $= \hat{y} = 1.2$
actual price $= y = 1.3$
} error by model $= 0.1$

5) compute loss function
↓
used for model performance evaluation

Loss Function

- used to evaluate model performance

eg.

area	age	price	K = 3 pred price	K = 5 pred price
1.2	0.5	0.8	0.8	1
1.3	0.5	0.9	0.95	1
1.4	1	1	0.9	1.2

model with less error → model with more accuracy
→ better model

- Loss function aggregates and quantifies the difference between actual target values and model predicted values.

- It quantifies the errors/ mistakes the model has made.

For regression models there are 3 widely used loss functions

1) Mean Absolute Error MAE
2) Mean Squared Error MSE
3) Root Mean Squared Error RMSE

Mean Absolute Error

eg.

Actual value	K=3 pred value	K=5 pred value
12	11	14
15	14	17
20	21	22

MAE = avg of absolute errors

$$MAE(K=3) = \frac{|12-11| + |15-14| + |20-21|}{3}$$

$$= \frac{1+1+1}{3} = 1$$

$$MAE(K=5) = \frac{|12-14| + |15-17| + |20-22|}{3}$$

$$= 2$$

here

$$MAE(K=3) < MAE(K=5)$$

lesser error $\longrightarrow$ better model

hence $K=3$ is better model than $K=5$

MAE formula

actual values $= Y = [Y_1 \quad Y_2 \quad Y_3 \ldots Y_n]$

predicted values $= \hat{Y} = [\hat{Y}_1 \quad \hat{Y}_2 \quad \hat{Y}_3 \ldots \hat{Y}_n]$

$$MAE = \frac{\sum_{i=1}^{n} |Y_i - \hat{Y}_i|}{n}$$

Mean Squared error

mse = avg of squares of differences between actual values and predicted values

$$MSE = \frac{\sum_{i=1}^{n} (Y_i - \hat{Y}_i)^2}{n}$$

$$MSE(K=3) = \frac{(12-11)^2 + (15-14)^2 + (20-21)^2}{3} = 1$$

$$MSE(K=5) = \frac{(12-14)^2 + (15-17)^2 + (20-22)^2}{3} = 4$$

$mse(K=3) < mse(K=5)$

based on mse, $K=3$ is better than $K=5$

Root Mean Squared Error

$$RMSE = \sqrt{MSE}$$

$rmse(K=3) = \sqrt{1} = 1$

$rmse(K=5) = \sqrt{4} = 2$

$rmse(K=3) < rmse(K=5)$